White Paper

Metabolica™ Medical Food: A Metabolically Targeted Nutritional Support Formula for Cancer Patients

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Abstract

Background: Conventional oral nutrition supplements used in oncology were formulated decades ago for general malnutrition. They are built on sugar, maltodextrin, dairy protein and omega-6 seed oils, inputs that acutely raise glucose, insulin and insulin-like growth factor 1 (IGF-1), activate mTOR and inflammatory signaling, and supply the substrates on which tumor metabolism preferentially depends. **Objective**: To set out the metabolic rationale, formulation logic, comparative profile and clinical use of Metabolica™, a medical food designed to nourish healthy tissue while withholding growth-promoting substrates during cancer treatment and recovery. **Design**: Six coordinated metabolic levers were defined from the biology of tumor metabolism: (1) glucose restriction and insulin/IGF-1 axis modulation; (2) glutamine and protein modulation; (3) fat- and ketone-based energy; (4) inflammation and redox control; (5) physiologic micronutrient engineering; and (6) digestive tolerance. Every ingredient was selected against these levers and screened for nutrient–drug interactions, radiation synergy or antagonism, reactive oxygen species (ROS) modulation and effects on hepatic and renal clearance. **Results**: The formula combines a dual plant-protein system (rice and pea, approximately 15 g per serving) with lower leucine, lysine, methionine and glutamine content; medium-chain triglycerides (C8/C10) and β-hydroxybutyrate salts as non-glycemic fuel; agave inulin and monk fruit for near-zero glycemic load; cinnamon, ginger and deglycyrrhizinated licorice as bioactives; a digestive enzyme blend; and micronutrients at physiologic doses, with no added iron or copper and vitamin B12 held at 2.4 µg. Against Ensure® Plus and Boost® High Protein, Metabolica™ delivers 0 g sugar, no dairy, no omega-6 oils and controlled trace metals. **Conclusions**: Metabolica™ represents a paradigm shift in therapeutic nutrition, from calorie replacement to metabolically selective support. It nourishes the patient across chemotherapy, radiation, surgery and remission while limiting activation of pro-growth and inflammatory pathways.

Keywords: medical food; metabolic oncology; insulin-like growth factor 1; mTOR; AMPK; ketone metabolism; β-hydroxybutyrate; medium-chain triglycerides; glutamine; plant protein; oncology nutrition

Medical food · Patent pending

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1. Introduction

Cancer is both a genetic and a metabolic disease, characterized by disrupted energy regulation and mitochondrial inefficiency. Tumor cells display metabolic inflexibility, relying on glucose and glutamine to fuel growth while avoiding oxidative phosphorylation, a hallmark known as the Warburg effect [9]. That inflexibility offers a therapeutic window: by modulating nutrient availability and energy sources, nutrition can influence cancer metabolism.

Most commercial meal replacements ignore this biology. Products such as Ensure® Plus and Boost® High Protein were designed for general malnutrition and recovery, not for the altered metabolic landscape of cancer. Their sugar, maltodextrin, dairy protein and omega-6 oils

acutely raise glucose and insulin, activate the IGF-1/mTOR axis, enhance glycolytic metabolism and increase systemic inflammation [4,5,23]. The result is a class of products that provide general-purpose caloric and micronutrient support, without regard to the metabolic context of cancer, while also supplying the substrates on which tumor metabolism preferentially depends.

Metabolica™, in contrast, was formulated to address the unique metabolic vulnerabilities of cancer patients undergoing treatment or recovery: its macronutrient ratio is designed to nourish and protect normal tissue while avoiding stimulation of growth pathways driven by glucose, glutamine and IGF-1, and its micronutrients are dosed specifically to support these metabolic objectives. The formulation aligns with the principles of metabolic oncology and works on six coordinated levers rather than one. It targets the two dominant fuel sources for proliferative tumor metabolism, glucose and glutamine, and moderates insulin/IGF-1 and mTOR growth signaling, while supporting cellular resilience through fat- and ketone-based energy [8,9]. It removes the omega-6 oils and added iron and copper that sustain inflammatory and oxidative signaling [2,22,23], keeps micronutrients within physiologic range, and includes digestive enzymes and soothing botanicals so that it is tolerated in a treatment-affected gut.

This white paper describes the mechanistic targets behind the formula (Section 2), compares it with the leading meal replacements (Section 3), sets out use protocols across the stages of cancer care (Section 4), reviews the supporting science (Section 5) and summarizes safety and clinical positioning (Section 6).

2. Metabolic Rationale and Mechanistic Targets

2.1. Six Coordinated Metabolic Levers

Metabolica™ is built around six coordinated metabolic levers, pulled at once rather than one at a time (Figure 1). Each lever addresses a specific way in which cancer metabolism diverges from the metabolism of healthy tissue. The first four are signaling and fuel pathways, shown in Sections 2.2 to 2.5 as pathway models; the remaining two shape how the formula behaves inside the body and in a treatment-affected gut.

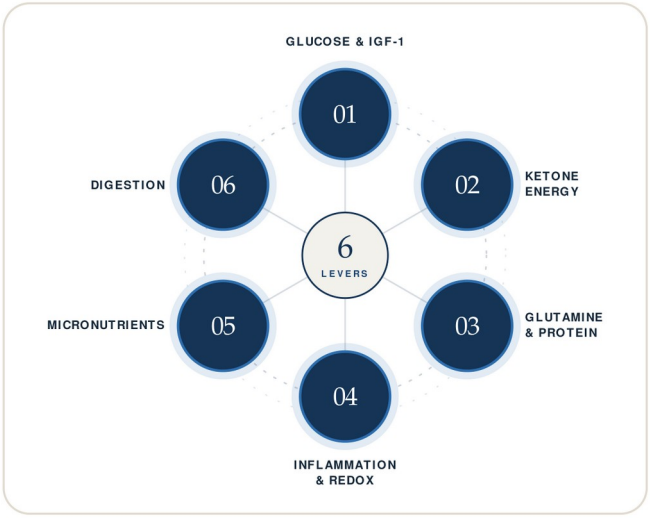


Figure 1. The six coordinated metabolic levers of the Metabolica™ formulation. (01) Glucose restriction and insulin/IGF-1 axis modulation; (02) fat- and ketone-based cellular energy; (03) glutamine and protein modulation; (04) inflammation and redox control; (05) physiologic micronutrient engineering; (06) digestive tolerance.

2.2. Glucose Restriction and Insulin–IGF-1 Axis Modulation

Why this matters

Most commercial meal replacements rely on maltodextrin, sugar or dairy proteins as major ingredients. These ingredients acutely raise glucose and insulin, activating the IGF-1/mTOR axis that promotes anabolic cell growth and suppresses autophagy [4,5]. Chronic activation of these pathways correlates with tumor proliferation and reduced treatment response. Figure 2 traces the cascade from dietary input through the insulin and IGF-1 receptors, PI3K, AKT and mTORC1 to its downstream outputs, and contrasts the signaling state produced by a conventional shake with the state produced by Metabolica™.

Metabolica™'s approach

Metabolica™ removes all simple sugars and maltodextrin, relying instead on:

- **Organic agave inulin**, a low-glycemic prebiotic fiber (glycemic index approximately 1) that supports microbiome balance and short-chain fatty acid (SCFA) production [6,25].
- **Monk fruit extract**, a zero-glycemic natural sweetener that does not stimulate insulin secretion and exhibits potential anti-inflammatory properties [7].
- **Medium-chain triglycerides (MCTs) and β-hydroxybutyrate (BHB) salts**, which provide non-glycemic, ketone-based energy that bypasses glycolytic pathways [8].
- **Magnesium and zinc**, cofactors that support healthy insulin sensitivity.

Result: This composition minimizes insulinogenic signaling, creating a metabolic environment supportive of healthy cellular efficiency rather than tumor growth.

2.3. Glutamine and Protein Modulation

Why this matters

Glutamine acts as a secondary fuel for many tumors, feeding the tricarboxylic acid (TCA) cycle through glutaminolysis while supplying nitrogen for nucleotide synthesis [9]. Animal proteins, high in leucine and lysine, are potent mTOR activators [10]. Ensure® and Boost® rely on milk proteins (casein, whey) that elevate serum IGF-1 and supply abundant branched-chain amino acids; in cancer patients this may inadvertently support a pro-growth state [11]. Methionine dependence is a further, well-documented feature of many tumors, and dietary methionine restriction has been shown to prime anti-tumor immunity [26–29]. Figure 3 contrasts glutamine flux and anabolic drive from a dairy protein source with that from the rice and pea blend.

Metabolica™'s approach

- **Dual plant-protein system (rice + pea).** The blend supplies all essential amino acids with lower leucine, lysine and methionine content than dairy, reducing anabolic drive while preventing muscle wasting [12,13].
- **Moderate total protein (approximately 15 g per serving),** adequate for tissue maintenance but below the mTOR-activation threshold.
- **A protein-to-fat ratio that prioritizes preservation, not hypertrophy.**
- **Digestive enzymes.** Protease, lipase and carbohydrase enhance digestion and tolerance in compromised gastrointestinal tracts.

Result: Support of lean tissue integrity without fueling tumor-associated nitrogen metabolism.

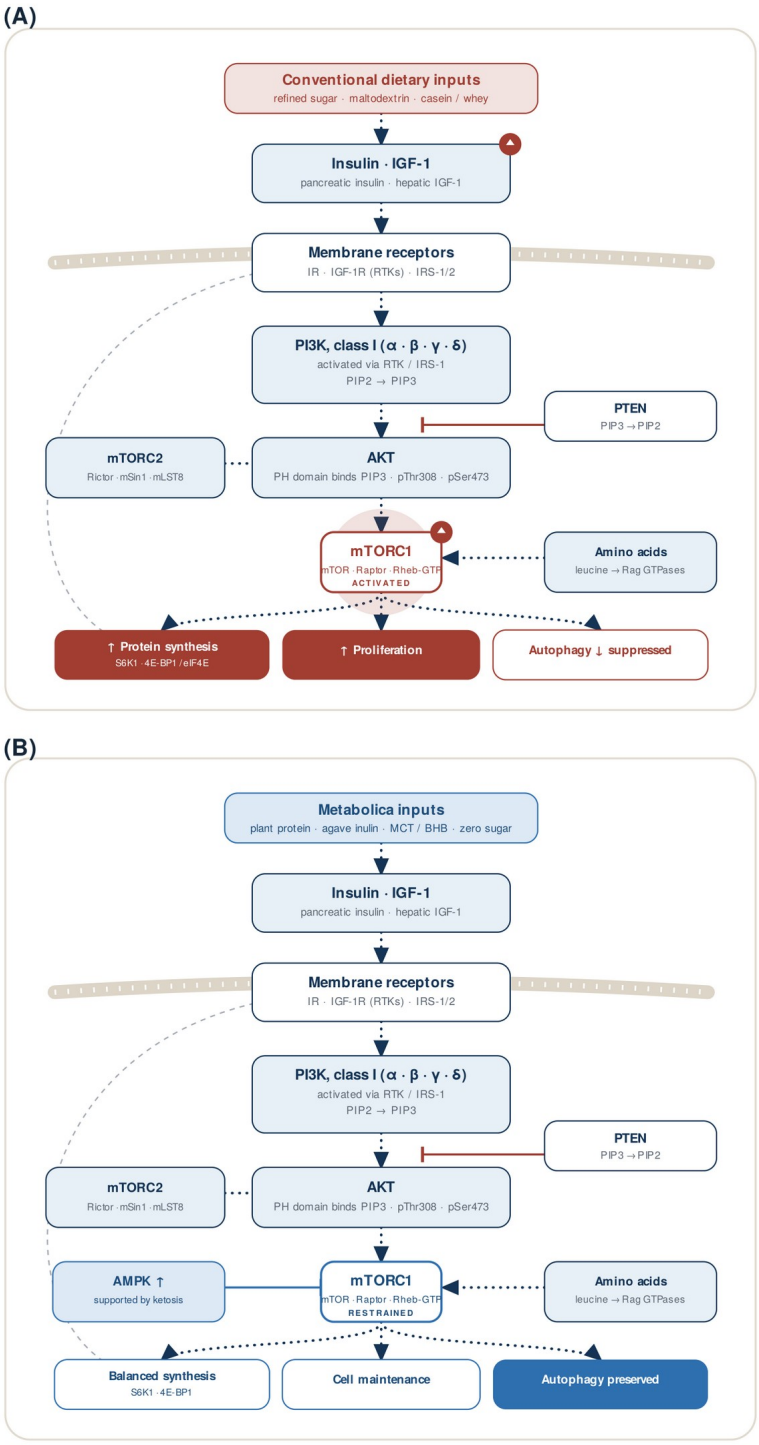


Figure 2. Insulin–IGF-1–PI3K–AKT–mTOR growth signaling under two nutritional inputs. (**A**) Conventional inputs (refined sugar, maltodextrin, casein and whey) raise insulin and IGF-1, which engage their receptors and IRS-1/2, activate class I PI3K (PIP2 → PIP3) and AKT, and drive mTORC1 into an activated state: increased protein synthesis (S6K1, 4E-BP1/eIF4E), increased proliferation and suppressed autophagy. (**B**) Metabolica™ inputs (plant protein, agave inulin, MCT/BHB, zero sugar) keep insulin and IGF-1 low, moderate leucine delivery to mTORC1 via Rag GTPases, and support AMPK activity through nutritional ketosis, which restrains mTORC1; outputs shift to balanced synthesis, cell maintenance and preserved autophagy. PTEN opposes PI3K (PIP3 → PIP2); the dashed line is S6K1 negative feedback. Solid nodes are driven outputs; outlined nodes are supported or restrained states.

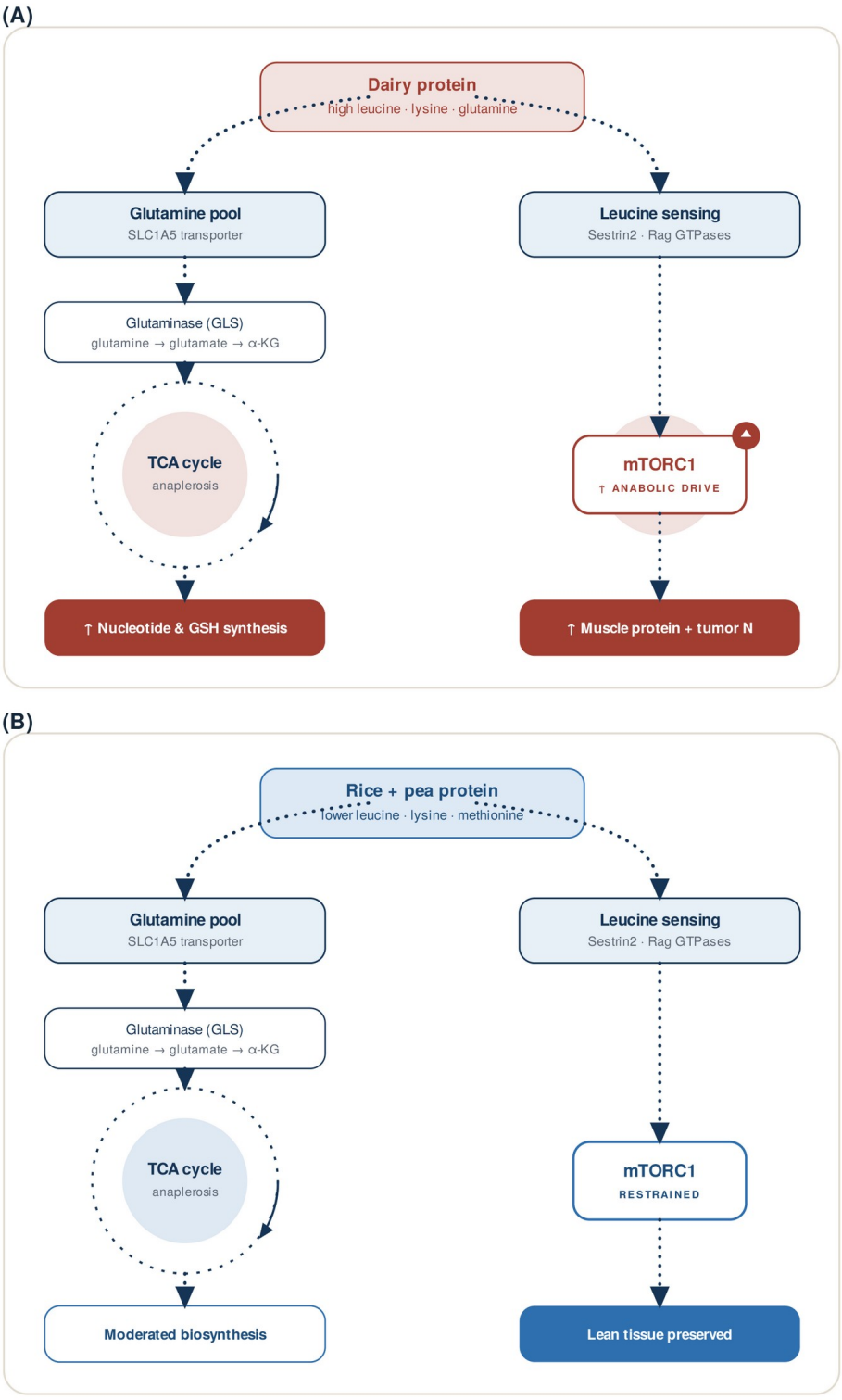


Figure 3. Glutamine flux and anabolic drive by protein source. (**A**) Dairy protein, high in leucine, lysine and glutamine, expands the glutamine pool (SLC1A5 transport) and feeds glutaminase (GLS), converting glutamine to glutamate and α-ketoglutarate for TCA anaplerosis, with increased nucleotide and glutathione synthesis; in parallel, leucine sensing through Sestrin2 and Rag GTPases activates mTORC1 and increases both muscle protein synthesis and tumor nitrogen supply. (**B**) The rice and pea protein blend delivers complete amino acids with lower leucine, lysine and glutamine, moderating flux through GLS and holding mTORC1 below its activation threshold, so that lean tissue is preserved without excess anabolic drive.

2.4. Lipid and Ketone-Based Metabolism

Why this matters

Fatty-acid oxidation and ketone utilization provide efficient, low-inflammatory energy for healthy cells while depriving many tumors of their preferred glycolytic substrates. Ketones such as BHB exhibit anti-inflammatory and anti-proliferative effects [8,14,15]. Healthy cells remain metabolically flexible and oxidize fat and ketones efficiently through the mitochondria, whereas many tumor cells rely on aerobic glycolysis and express little of the enzyme (BDH1) needed to use ketones (Figure 4).

Metabolica™'s approach

- **MCTs (C8/C10) and BHB salts** (sodium, potassium, magnesium and calcium forms) promote mitochondrial efficiency and are rapidly oxidized through β-oxidation [16].
- **Lower oxidative stress and systemic inflammation** through a higher NAD^+/NADH ratio and reduced ROS generation.
- **Mild, physiologic nutritional ketosis** that enhances metabolic flexibility without requiring strict carbohydrate restriction.

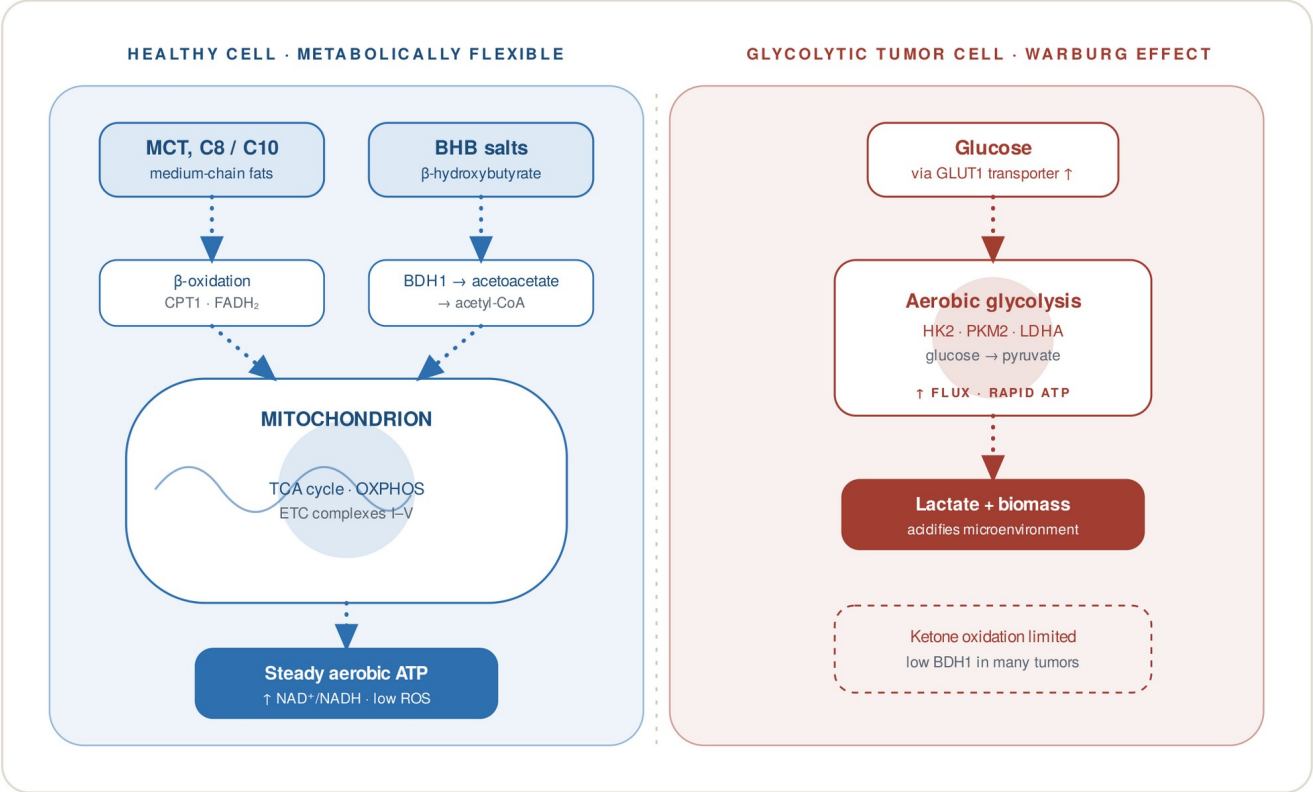


Figure 4. Fueling healthy cells with ketones rather than glucose. Left: in a metabolically flexible healthy cell, medium-chain fats (C8/C10) enter β-oxidation (CPT1, $FADH_2$) and BHB is converted by BDH1 to acetoacetate and acetyl-CoA; both feed the mitochondrial TCA cycle and oxidative phosphorylation (electron transport chain complexes I–V), yielding steady aerobic ATP with a high NAD^+/NADH ratio and low ROS. Right: a glycolytic tumor cell imports glucose through up-regulated GLUT1 and runs aerobic glycolysis (HK2, PKM2, LDHA) at high flux, exporting lactate and biomass that acidify the microenvironment; ketone oxidation is limited by low BDH1 expression in many tumors.

2.5. Inflammation and Redox Modulation

Why this matters

Omega-6 dominant seed oils raise arachidonic acid and prostaglandin E_2 (PGE_2), while excess iron and copper drive Fenton-reaction free radicals [2,22,23]. Together they sustain nuclear factor κB (NF-κB) signaling and a pro-inflammatory, oxidative environment that promotes angiogenesis and immune suppression (Figure 5).

Metabolica™'s approach

- **MCT only, no omega-6 seed oils,** reducing the arachidonic acid and PGE_2 load.
- **Cinnamon extract**, which enhances insulin receptor sensitivity [18].
- **Ginger root extract**, which reduces interleukin-6 (IL-6) and tumor necrosis factor α (TNF-α) and alleviates nausea [19,20].
- **Deglycyrrhizinated licorice (DGL)**, which soothes gastric mucosa and has been studied in radiation-induced mucositis [21].
- **No added iron or copper,** limiting Fenton-driven ROS; **zinc, selenium and manganese** as cofactors for superoxide dismutase (SOD) and glutathione peroxidase (GPx); **BHB** for NLRP3 inflammasome modulation [17].

2.6. Micronutrient Balance and Physiologic Dosing

Metabolica™'s micronutrient engineering extends far beyond simple daily value considerations. Cancer biology demonstrates that certain trace minerals can function as growth accelerants. Copper is a known cofactor in angiogenesis and tumor invasion [1], while iron can amplify oxidative stress and feed tumor proliferation [2,22]; both were intentionally excluded from the formula. Conversely, micronutrients such as vitamin B12, zinc and selenium are essential for host immune function, redox balance and DNA repair, but can stimulate growth when provided in excess [3]. Metabolica™ therefore delivers physiologic levels, not pharmacologic doses:

- Iron held low, approximately 0.8 mg, with no added copper.
- Vitamin B12 at a physiologic 2.4 μg.
- Magnesium, zinc, selenium and manganese at optimal cofactor levels for antioxidant and DNA-repair enzymes.
- Antioxidant vitamins balanced at approximately 50% of daily value, avoiding the high-dose antioxidant exposure that is generally discouraged during active treatment.

Every micronutrient and botanical was further evaluated for nutrient–drug interactions, radiation synergy or antagonism, ROS modulation and effects on hepatic and renal clearance, to ensure compatibility with chemotherapy, radiation, immunotherapy and peri-operative protocols. This includes exclusion of compounds known to interfere with drug metabolism at clinically relevant doses (for example, cytochrome P450 modulators).

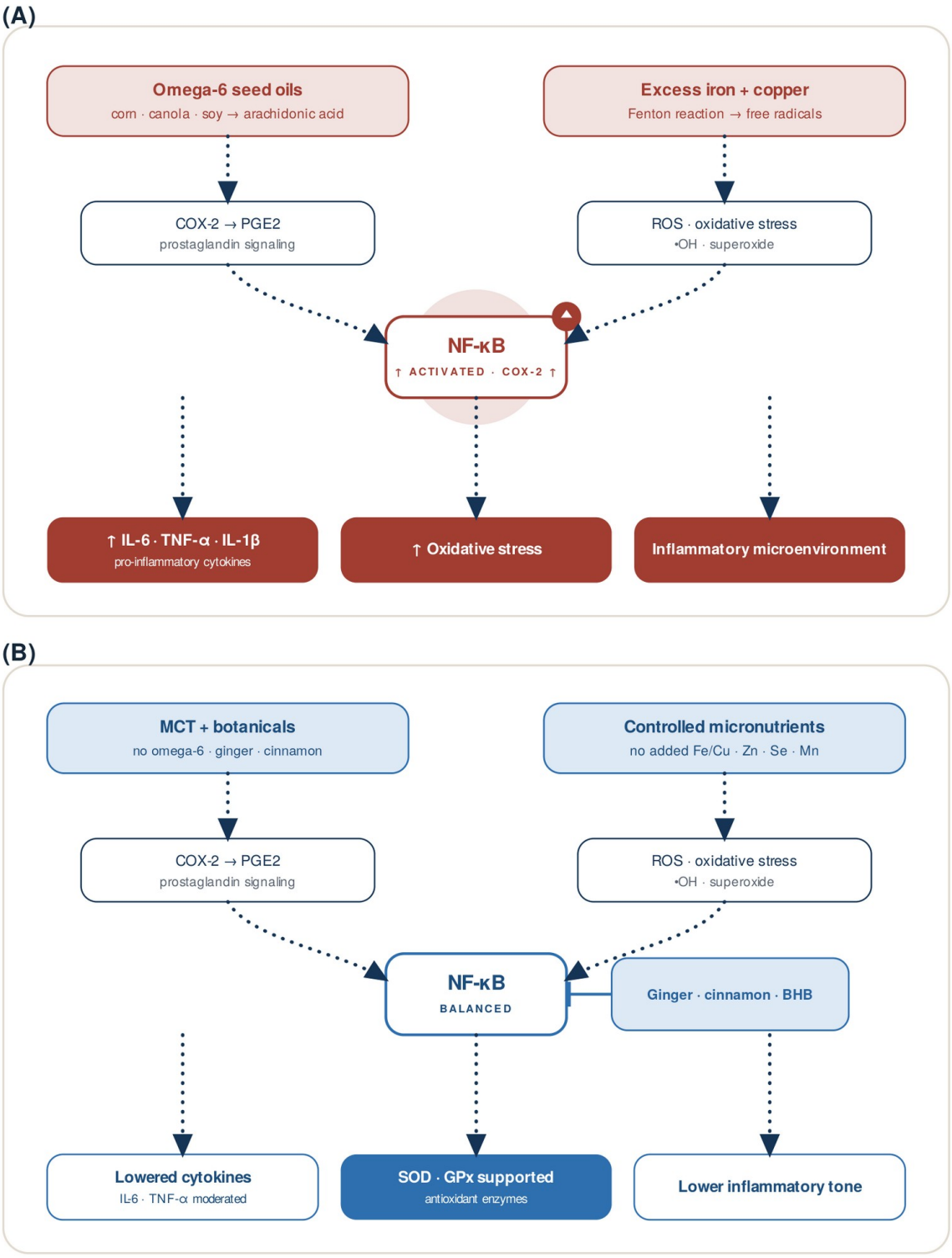


Figure 5. Inflammatory and oxidative signaling under two formulation strategies. (**A**) Omega-6 seed oils (corn, canola, soy) supply arachidonic acid to COX-2, generating PGE_2, while excess iron and copper drive Fenton-reaction hydroxyl radical and superoxide formation; both converge on NF-κB, which up-regulates COX-2 and the pro-inflammatory cytokines IL-6, TNF-α and IL-1β, producing oxidative stress and an inflammatory microenvironment. (**B**) Metabolica™ removes the omega-6 and Fe/Cu inputs, and ginger, cinnamon and BHB support healthy NF-κB and COX-2 activity, calming the cascade. The ⊣ symbol indicates botanical support acting against a node.

2.7. Digestive Support and Patient Tolerance

Because cancer therapies target rapidly dividing cells, including the enterocytes that line the gastrointestinal tract, digestive disturbances are common during treatment. A metabolic support formula intended for this patient population must therefore also support gastrointestinal function and tolerance. To that end, Metabolica™ includes:

- A **digestive enzyme blend** (protease, lipase, carbohydrase) for improved macronutrient absorption.
- **Ginger and DGL** for soothing of the gastric mucosa and antiemetic support [19–21].
- **Agave inulin** as a gentle prebiotic fiber for microbiome health [6,25].

The selected fiber, paired with monk fruit, cinnamon and natural vanilla, creates a pleasant flavor profile and balanced sweetness while maintaining strict glycemic control. These elements enhance tolerability compared with lactose- or maltodextrin-containing formulas. Table 1 summarizes the six targets, the conventional risk factor at each and the corresponding Metabolica™ strategy.

Table 1. Mechanistic overview of the Metabolica™ formulation by metabolic target.

Metabolic Target	Conventional Risk Factor	Metabolica™ Strategy	Primary Mechanism
IGF-1 / mTOR	Milk proteins, leucine excess	Plant proteins low in lysine and methionine	↓ Anabolic signaling
Glycolysis / Warburg effect	Sugars, maltodextrin	Low-carbohydrate design, MCT, BHB	↑ Ketone metabolism
Inflammation	Omega-6 oils	MCT + ginger	↓ NF-κB activity
Oxidative stress	Iron, copper, B12 excess	Controlled micronutrients; no added iron or copper	↓ ROS generation
Digestive tolerance	Casein, soy isolates, emulsifiers	Enzymes + DGL + ginger; inulin fiber	↑ GI comfort and absorption

3. Comparative Analysis: Metabolica™ versus Leading Meal Replacements

Most commercial meal replacements for cancer or general nutrition are formulated around calorie density and palatability, not metabolic compatibility. Popular products such as Ensure® Plus and Boost® High Protein were designed decades ago for general malnutrition and recovery, not for the altered metabolic landscape of cancer. These conventional formulas can inadvertently activate insulin/IGF-1 and mTOR pathways, enhance glycolytic metabolism and increase systemic inflammation, conditions that may undermine metabolic stability during therapy. Metabolica™ was developed as a metabolically aligned alternative, providing energy and nutrition that support the host rather than the tumor microenvironment.

3.1. Comparative Ingredient Overview

Table 2 sets the first five listed ingredients, protein source, sugar content and fat source of each product side by side, with the metabolic considerations that follow from them.

Table 2. Comparative ingredient overview of Ensure® Plus, Boost® High Protein and Metabolica™.

Product	First Five Ingredients	Primary Protein	Sugar (g/serving)	Fat Source	Metabolic Considerations (Scientific Basis)
Ensure® Plus	Water, corn maltodextrin, sugar, milk protein concentrate, canola oil	Casein / whey	22 g	Canola and corn oils	High glycemic load; dairy proteins increase IGF-1/mTOR; omega-6 fats promote inflammation [4,5,23]
Boost® High Protein	Water, glucose syrup, milk protein concentrate, sugar, soy oil	Casein / whey	20 g	Soy and canola oils	Rapid insulin/IGF-1 response; soy oil omega-6 dominance; dairy proteins stimulate anabolic signaling [11]
Metabolica™	Rice protein, MCT (C8/C10), organic pea protein, agave inulin, β-hydroxybutyrate salts	Rice + pea	0 g	MCT (C8/C10)	Low glycemic; supports ketone metabolism; anti-inflammatory fat profile; no dairy, sugar or maltodextrin [8,12,16]

3.2. Glycemic and Insulinogenic Load

High-carbohydrate shakes depend on maltodextrin or glucose syrup, both with glycemic indices above 85. These rapidly elevate glucose and insulin, activating mTOR and accelerating cellular proliferation [4,9]. Metabolica™ employs a different energy paradigm: agave inulin (glycemic index approximately 1) stabilizes glycemia and supports beneficial gut flora [6]; monk fruit extract supplies zero-glycemic natural sweetness [7]; and BHB salts provide an alternate oxidative fuel [8]. This yields a near-zero glycemic load while maintaining steady-state energy production.

3.3. Protein Source and Anabolic Signaling

Dairy-based proteins (casein, whey) are among the most anabolic because of their leucine and lysine content. While beneficial for athletic populations, in oncology they may up-regulate IGF-1 and mTOR activity, promoting anabolic signaling [10,11]. Metabolica™'s rice and pea blend supplies complete amino acids with lower lysine and methionine, limiting anabolic drive [12,13]; moderate protein (approximately 15 g per serving) supports tissue integrity without excess nitrogen load; and digestive enzymes ensure efficient utilization and reduce gastrointestinal distress.

Result: Maintenance of lean mass without promoting tumor-favorable metabolism.

3.4. Lipid Composition and Inflammation

Most commercial formulas use canola, soy or corn oils, which are rich in omega-6 linoleic acid, the precursor of PGE_2 and inflammatory cytokines [23]. Metabolica™ uses medium-chain triglycerides exclusively as its fat source. MCTs are rapidly absorbed and converted to ketones, do not contribute to adipose storage and exhibit anti-inflammatory and mitochondrial-supportive effects [15,16]. This lipid profile supports sustained cellular energy while minimizing systemic inflammation.

3.5. Micronutrient Density and Physiologic Dosing

Table 3 contrasts the trace-mineral and vitamin dosing of conventional formulas with the Metabolica™ design. The balance avoids the nutrient excesses that can perturb redox homeostasis or stimulate proliferation.

Table 3. Micronutrient dosing: conventional formulas versus the Metabolica™ design.

Parameter	Conventional Formulas (Ensure/Boost)	Metabolica™ Design
Iron	8–10 mg	0.8 mg; avoids iron-driven oxidative stress [2]
Vitamin B12	6–12 µg	2.4 µg; physiologic level, avoids angiogenic signaling [3]
Copper	0.8–1.0 mg	Not added; prevents copper-mediated angiogenesis [1]
Antioxidant vitamins	Often supraphysiologic	Balanced at approximately 50% DV; supports antioxidant systems without redox imbalance
Selenium and zinc	Variable	Physiologic (11 µg Se, 5 mg Zn); maintains enzymatic antioxidant defense

3.6. Digestive Tolerance and Absorption

Digestive intolerance, including bloating, diarrhea and nausea, is a frequent complaint with conventional formulas; lactose, soy isolates and emulsifiers are common irritants. Metabolica™ mitigates these issues through its digestive enzyme blend (protease, lipase, carbohydrase), ginger and DGL extracts for gastric soothing and anti-nausea effects [19–21], and inulin prebiotic fiber for microbiome support [6]. This ensures consistent absorption and comfort even during chemotherapy or in gastrointestinally sensitive states.

3.7. Summary Comparison

Table 4. Summary comparison of energy source, protein source, glycemic profile and formulation features.

Parameter	Ensure® Plus	Boost® High Protein	Metabolica™
Energy source	Sugar + maltodextrin	Glucose + sugar	MCT + BHB + inulin
Protein source	Casein / whey	Casein / whey	Plant (rice + pea)
Glycemic index	High (>70)	High (>65)	Very low (<15)
Sugar (g)	22	20	0
Dairy free	✗	✗	✓
Ketogenic support	✗	✗	✓
Inflammatory oils	Canola / corn	Soy / canola	None (MCT only)
Iron / copper added	Yes	Yes	Controlled / excluded
Digestive enzymes	None	None	Included

3.8. Interpretive Summary

Metabolica™ represents a paradigm shift in therapeutic nutrition: from formulas that deliver calories without distinguishing between the patient and the tumor, to a formula that is metabolically selective. Its low-glycemic, plant-protein and fat-centric design provides targeted nutrition to healthy tissue, which remains metabolically flexible and uses fat and ketones efficiently, while restricting the glucose, glutamine and growth-factor signaling on which tumor metabolism preferentially depends. The formula is thereby built around the metabolic vulnerabilities that distinguish cancer biology from normal physiology, consistent with evidence that moderating insulin/IGF-1, glutamine flux and inflammatory signaling can improve metabolic resilience during cancer care. By eliminating sugars, dairy proteins and omega-6 oils, Metabolica™ delivers nutritionally complete support for the patient while withholding the substrates that conventional formulas supply to the tumor.

4. Clinical Applications and Use Protocols

Metabolica™ supports patients across all phases of cancer care, from active treatment to surgical recovery and long-term remission. Each stage presents unique metabolic challenges, including mitochondrial fatigue, inflammation, insulin resistance and digestive intolerance. Unlike conventional high-sugar or dairy-based formulas, Metabolica™ provides targeted metabolic support through balanced macronutrients and physiologic micronutrient dosing, maintaining energy stability without stimulating growth signaling pathways. It is a flexible nutritional adjunct rather than a rigid protocol: the metabolic challenge changes across treatment and recovery, and so does the way the formula is used.

4.1. During Chemotherapy or Radiation

Metabolic context

Chemotherapy and radiation impose oxidative and metabolic stress, elevating inflammatory cytokines (IL-6, TNF-α) and disrupting glucose tolerance. High-carbohydrate nutritional formulas exacerbate hyperglycemia and IGF-1/mTOR activation, worsening fatigue and inflammation.

Metabolica™'s supportive role

- MCTs and BHB salts provide efficient, non-glycemic energy for healthy cells.
- Plant proteins (rice + pea) preserve lean tissue with low glutamine content.
- Physiologic micronutrients (Mg, Se, Zn) enhance mitochondrial enzymes and antioxidant defenses.
- Cinnamon supports insulin sensitivity and glycemic stability [18].
- Ginger and DGL mitigate nausea and improve tolerance [19–21].

Recommended use

- **Serving:** 2–4 servings daily (300–600 kcal).
- **Timing:** Mid-morning or mid-afternoon between meals, or when appetite is reduced.
- **Preparation:** Mix with water or unsweetened nut milk; may be blended with small amounts of low-glycemic fruit (for example, berries).

> **Clinical note.** Metabolica™ may serve as a partial meal replacement on treatment days when solid intake is limited.

4.2. Pre-Surgical and Peri-Operative Nutrition

Metabolic context

Surgical stress induces transient insulin resistance, catabolism and inflammatory activation. Optimizing peri-operative nutrition improves wound healing and recovery [24].

Metabolica™'s supportive role

- Maintains euglycemia and minimizes peri-operative hyperglycemia.
- Supplies rapidly absorbable MCT energy without digestive burden.
- Provides balanced amino acids for tissue integrity without mTOR overstimulation.
- Delivers micronutrients (vitamins A, D, E and K; zinc; selenium) for immune competence and collagen synthesis.

Recommended use

- **Pre-surgery (5–7 days prior):** 1–2 servings daily with meals to optimize stores.
- **Post-surgery:** 2–4 servings daily as tolerated for recovery and appetite support.

> **Clinical note.** Free of dairy and maltodextrin, Metabolica™ reduces post-operative bloating and nausea.

4.3. Recovery and Remission Support

Metabolic context

After treatment, patients frequently experience mitochondrial inefficiency, fatigue and sarcopenia. Long-term goals focus on restoring metabolic flexibility while maintaining low-inflammatory and low-IGF-1 states.

Metabolica™'s supportive role

- Ketone-based energy enhances endurance and cognitive clarity.
- Low-glutamine, plant-based proteins maintain muscle without anabolic excess.
- Inulin fiber restores microbiome balance and gut–brain signaling [6,25].
- Physiologic micronutrients prevent both deficiency and toxic accumulation.

These elements collectively sustain remission by supporting balanced mitochondrial metabolism [9,15].

Recommended use

- **Maintenance:** 1–2 servings daily between meals or as a meal replacement.
- **Enhanced energy:** Blend with extra-virgin organic olive oil or other targeted functional foods for higher caloric density.

4.4. Digestive and Tolerance Considerations

Digestive issues such as bloating, nausea and altered motility are frequent during therapy. Metabolica™ supports gastrointestinal function through its digestive enzyme blend (improved macronutrient absorption), ginger and DGL extracts (reduced mucosal irritation and nausea), inulin prebiotic (support for *Bifidobacterium* and *Lactobacillus* populations) and the absence of

lactose, gluten, soy and emulsifiers (fewer intolerance reactions). Table 5 summarizes the metabolic challenge, mechanism and clinical goal at each stage of care.

Table 5. Clinical summary by stage of care.

Stage of Care	Metabolic Challenge	Metabolica™ Mechanism	Clinical Goal
Chemotherapy / radiation	Glucose dependency, mitochondrial fatigue	MCT + BHB → non-glycemic energy	Preserve energy and lean mass
Pre-surgery	Insulin resistance, inflammation	Low-glycemic, balanced amino acids	Optimize metabolic readiness
Post-surgery	Wound healing, GI intolerance	MCT + enzymes + micronutrients	Support recovery and gut function
Remission	Chronic inflammation, metabolic inflexibility	Ketones + balanced protein/fat	Maintain long-term metabolic health

Metabolica™ is thus a metabolically aligned nutritional adjunct for oncology and recovery care. Its low-glycemic, fat-forward formulation and physiologic micronutrient profile nourish the patient while avoiding substrates that feed tumor metabolism. This integrative design supports stable energy, improved tolerance and resilient recovery.

5. Scientific Discussion and Mechanistic Overview

5.1. IGF-1 and mTOR Signaling Modulation

The IGF-1/mTOR axis governs cellular growth and nutrient sensing. Elevated IGF-1, often driven by dairy and high animal-protein diets, activates mTOR, stimulating proliferation and angiogenesis while suppressing autophagy [4]. Metabolica™ avoids casein- and whey-induced IGF-1 elevations by excluding dairy proteins; maintains tissue repair without anabolic excess through moderate plant protein (approximately 15 g); and supports insulin sensitivity with magnesium and zinc (Figure 2).

Result: Controlled nutrient signaling that supports recovery without promoting unchecked cell growth.

5.2. AMPK Activation and Metabolic Flexibility

AMP-activated protein kinase (AMPK) regulates cellular energy status. When activated, AMPK promotes fat oxidation, mitochondrial biogenesis and autophagy while inhibiting mTOR. In Metabolica™, BHB salts increase the $NAD^+/NADH$ ratio, enhancing oxidative metabolism [14], and MCTs provide direct substrates for mitochondrial β-oxidation [16].

Outcome: Improved metabolic flexibility, allowing healthy cells to alternate between fat and glucose utilization and counteracting tumor glycolysis dependence.

5.3. Glycolysis and Glutaminolysis: Restricting Tumor Fuel Supply

Tumor cells rely on aerobic glycolysis and glutaminolysis for rapid ATP and biosynthetic precursors; high glucose and glutamine availability sustain this state (Figures 3 and 4). Metabolica™ reduces the glycolytic substrate load through zero added sugars and maltodextrin; moderates glutamine flux through plant proteins low in lysine and methionine [13,26]; and

supplies BHB salts as an alternative oxidative fuel for normal cells that is poorly metabolized by many tumors [8,15].

Outcome: Controlled fuel availability that supports normal tissues but constrains tumor energetics.

5.4. Oxidative Stress, Inflammation and the Tumor Microenvironment

Reactive oxygen species and pro-inflammatory cytokines (IL-6, TNF-α, C-reactive protein) promote angiogenesis and immune suppression. Omega-6 dominant oils (corn, canola, soy) elevate PGE_2, fueling chronic inflammation [23] (Figure 5). In Metabolica™, MCTs reduce mitochondrial ROS formation; ginger and cinnamon suppress NF-κB and COX-2 activity [18,20]; zinc, selenium and manganese support the antioxidant enzymes SOD and GPx; and balanced vitamins A, C and E prevent pro-oxidative excess.

Outcome: Lower inflammatory signaling, improved mitochondrial resilience and protection of normal tissue during therapy.

5.5. Ketone Biology and Cellular Signaling

Ketones are not merely energy substrates; they exert epigenetic and anti-inflammatory effects. Histone deacetylase (HDAC) inhibition by BHB up-regulates stress-resistance genes [14]; NLRP3 inflammasome inhibition by BHB reduces systemic inflammation [17]; and neuroprotective effects improve cognition and fatigue resistance during treatment. Metabolica™'s BHB complex (Na^+, K^+, Mg^{2+} and Ca^{2+} forms) provides mild, stable ketosis without requiring strict carbohydrate restriction, which is well suited to oncology and recovery settings.

5.6. Integrative Mechanistic Model

Table 6 integrates the pathways above, contrasting the response produced by conventional formulas with the response produced by Metabolica™ and the clinical implication of each.

Table 6. Integrative mechanistic model: pathway-level response to conventional formulas versus Metabolica™.

Pathway	Conventional Response (Ensure/Boost)	Metabolica™ Response	Clinical Implication
IGF-1 / mTOR	↑ via dairy proteins, sugars	↓ via plant protein, AMPK activation	Reduced anabolic drive
Glycolysis	↑ via maltodextrin / glucose	↓ via BHB and low-carbohydrate design	Less tumor substrate
AMPK	↓ with chronic overnutrition	↑ via ketosis	Improved metabolic flexibility
Inflammation	↑ via omega-6 oils	↓ via MCTs, ginger, DGL	Reduced cytokine load
Oxidative stress	↑ via iron, copper excess	↓ via controlled micronutrients	Protects DNA and mitochondria

5.7. Systems-Level Interpretation

Metabolica™ establishes a metabolically adaptive but growth-neutral state. Healthy cells utilize fats and ketones efficiently, while malignant cells, dependent on glucose and glutamine,

experience relative substrate restriction. This strategy enhances host metabolic resilience and immune competence without nutrient deprivation. Metabolica™ thus exemplifies terrain-directed nutrition: supporting the body's terrain rather than targeting the tumor itself.

6. Safety Profile and Clinical Positioning

6.1. Formulation Safety Principles

1. **Physiologic dosing.** Micronutrients are maintained at or near daily values, minimizing toxicity or interference with medical therapies.
2. **Non-glycemic energy sources.** No added sugars or maltodextrin; caloric energy is derived from fats and ketones.
3. **Allergen-conscious formulation.** Dairy-free, gluten-free, soy-free and lactose-free, with no artificial sweeteners or preservatives.
4. **Digestive support.** Enzymes and botanicals support tolerability even during chemotherapy-related mucositis or gastrointestinal upset.
5. **Metal regulation.** No added iron or copper and no excessive B12, reducing angiogenic and oxidative risks [1–3].

6.2. Observed Tolerability

Patient and clinician feedback to date indicate that Metabolica™ is well tolerated, with reports of improved satiety and sustained energy, decreased postprandial fatigue, reduced gastrointestinal discomfort (bloating, nausea) and better glucose stability in insulin-resistant individuals. No adverse reactions have been observed at recommended dosages. Caution is advised in patients with significant renal impairment or electrolyte imbalance because of the presence of exogenous ketone salts.

6.3. Intended Use and Clinical Integration

Metabolica™ is a nutritional adjunct, not a treatment. It is designed to support patients during chemotherapy, radiation or surgery; to provide low-glycemic, anti-inflammatory nutrition compatible with oncologic and metabolic care; and to maintain energy and tissue repair in survivors pursuing long-term remission support. It is not intended to diagnose, treat, cure or prevent disease, but to enhance nutritional and metabolic stability.

Metabolica™ can be incorporated into dietitian-directed oncology nutrition protocols, integrative or functional medicine care plans emphasizing metabolic control, and clinical trials or observational studies exploring metabolic nutrition in cancer care. Given its physiologic profile and absence of pharmacologic doses, it is suitable for combination with most

therapeutic regimens under clinician supervision. Table 7 summarizes the key clinical takeaways.

Table 7. Key clinical takeaways.

Clinical Goal	Limitations of Conventional Formulas	Metabolica™ Advantage
Maintain energy during therapy	Sugar-based shakes spike insulin and IGF-1	Fat- and ketone-based energy stabilizes metabolism
Preserve lean tissue	Animal proteins overstimulate mTOR	Moderate plant proteins maintain tissue without anabolic drive
Control inflammation	Omega-6 oils increase cytokine cascades	MCTs + ginger reduce NF-κB activity
Support digestion	Dairy, soy and emulsifiers cause GI upset	Enzymes + DGL + inulin improve absorption
Reduce oxidative stress	High iron, copper and B12 induce ROS	Controlled micronutrients protect mitochondrial integrity

7. Conclusions

Metabolica™ Medical Food represents a paradigm shift in therapeutic nutrition, from calorie replacement to metabolically selective support. Every nutrient serves a deliberate metabolic purpose: plant-based proteins (rice + pea) provide sufficient amino acids without excessive lysine, methionine or glutamine; MCTs and BHB salts replace glucose as the primary fuel, reducing insulin and IGF-1 signaling; agave inulin and monk fruit extract maintain glycemic control and microbiome health; cinnamon and ginger help regulate inflammation; and micronutrients are provided in physiologic rather than pharmacologic quantities to avoid overstimulating growth or oxidative pathways.

Metabolica™ applies the principles of metabolic oncology to medical nutrition. It reflects a research-driven, clinically informed approach to oncology nutrition, one that nourishes patients while minimizing activation of pro-growth and inflammatory pathways. By shifting the energy supply from glucose toward fat and ketones, Metabolica™ is designed to feed the patient, not the cancer. Its physiologic, digestible and evidence-based formulation sets a new standard for metabolically selective nutrition in supportive cancer care, recovery and survivorship.

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Conflicts of Interest: The author is the developer of Metabolica™ and holds financial interests in Metabolica Inc. and Rallis Wellness Inc.

Abbreviations: AMPK, AMP-activated protein kinase; BHB, β-hydroxybutyrate; COX-2, cyclooxygenase-2; DGL, deglycyrrhizinated licorice; DV, daily value; GI, gastrointestinal (or glycemic index, where stated); GLS, glutaminase; GPx, glutathione peroxidase; HDAC, histone deacetylase; IGF-1, insulin-like growth factor 1; IL, interleukin; MCT, medium-chain triglyceride; mTOR(C1/C2), mechanistic target of rapamycin (complex 1/2); NF-κB, nuclear factor κB; PGE_2, prostaglandin E_2; PI3K, phosphoinositide 3-kinase; ROS, reactive oxygen species; SCFA, short-chain fatty acid; SOD, superoxide dismutase; TCA, tricarboxylic acid; TNF-α, tumor necrosis factor α.

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Important. Metabolica™ is a medical food intended to provide nutritional support and is not a drug. It is not intended to diagnose, treat, cure or prevent cancer or any disease. The pathway diagrams in this document describe the scientific rationale behind the formula's design and are provided for educational purposes only. Metabolica™ should be used under the supervision of a physician as part of a comprehensive care plan. These statements have not been evaluated by the Food and Drug Administration. Ensure® and Boost® are registered trademarks of their respective owners; comparative ingredient information is drawn from publicly available product labels.

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